

2 Goal and Area of Application

Evaluating the binding of a compound to its putative target is an important step in drug discovery and development. Without this critical information the relationship between the small molecule, the protein target and modulation of disease biology cannot be adequately studied *in vitro* or *in vivo*¹. The cellular thermal shift assay (CETSA)² is one important method to monitor target engagement, which has many potential applications to the drug discovery and development pipeline³. This SOP outlines the use of CETSA for measuring target engagement for target validation, screening or structure-activity relationship studies. Users are referred to other guidelines, protocols, or key papers for traditional thermal shift assays or other biophysical methods, such as surface plasmon resonance (SPR) to measure drug–protein interactions. In addition, other methods to measure target engagement in living cells are sometimes preferred. For such studies users are referred to key papers for NanoBRET⁴ target engagements studies and other live-cell adaptations⁵. If the target of a compound is not known CETSA–coupled to mass spectrometry (MS) readout can be used for target identification. These studies come with several different formats⁶ and users are referred to key methods papers in the area, such as thermal proteome profiling (TPP)⁷, CETSA-MS and PISA⁸, as well as another REMEDI4ALL SOG for MS-based proteomics methods. Of note, CETSA is a patented technology and it is recommended to contact the patent holder, Pelago Bioscience (www.pelagobio.com), for more information.

3 Definitions and Abbreviations

SOG	Standard Operating Guideline
SOP	Standard Operating Procedure
T _{agg}	Thermal aggregation curve
T _m	Thermal melt curve
ITDRF	IsoThermal Dose Response Fingerprint
TPP	Thermally Proteome Profiling
CETSA	Cellular Thermal Shift Assay
PISA	Proteome Integral Solubility Alteration
BRET	Bioluminescence Resonance Energy Transfer

4 Method

4.1 General

The Cellular Thermal Shift Assay is a method to quantify the interaction of a small molecule with a protein target³. The underlying biophysical principle of CETSA is similar to a traditional thermal shift assay, where the interaction between a small molecule and a protein stabilizes the protein to thermal denaturation resulting in a shift in the melt curve (Figure 1). Importantly, unlike the traditional thermal shift assay, CETSA is performed in the complex environment of a living cell or in a cell lysate.

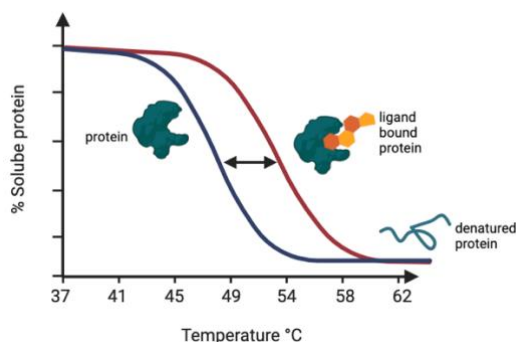


Figure 1. Schematic depicting the principle of CETSA. Created in BioRender. SEASHORE-LUDLOW, B. (2025)

<https://BioRender.com/rw3yjkb>

In CETSA the remaining soluble protein after a heat challenge is quantified to determine protein stabilization due to compound binding. This can in principle be accomplished using any method that simultaneously

measures protein abundance and can distinguish between soluble, natively folded protein and denatured, aggregated protein. For target validation using Western blot is likely the easiest method to establish. When multiple or parallelized experiments are necessary, microtiter plate compatible methods, such as AlphaLISA, are recommended. AlphaLISA is a bead-based, homogenous detection method, which is suitable to detect the level of correctly folded proteins at endogenous concentrations. In this setting, since two antibodies are used to recognize the protein, it is possible to selectively measure remaining soluble protein levels after the heat challenge used in CETSA. Alternative methods for adapting typical Western blot protocols to higher throughput have also been reported⁹.

There are two general formats for CETSA, one is thermal aggregation studies, where the melting of a protein is measured over a temperature series and compared to vehicle (Figure 2A). Here, the aggregation temperature (T_{agg}) can be determined. T_{agg} is used to differentiate between the equilibrium processes measured in traditional thermal shift assays, where T_m is measured, and the non-equilibrium processes measured in CETSA^{10,11}. Alternatively, one can perform isothermal dose response (ITDR) studies at a single temperature and a concentration series of the compounds of interest. Using this method target engagement between different small molecules or a series of analogues can be easily compared (Figure 2).

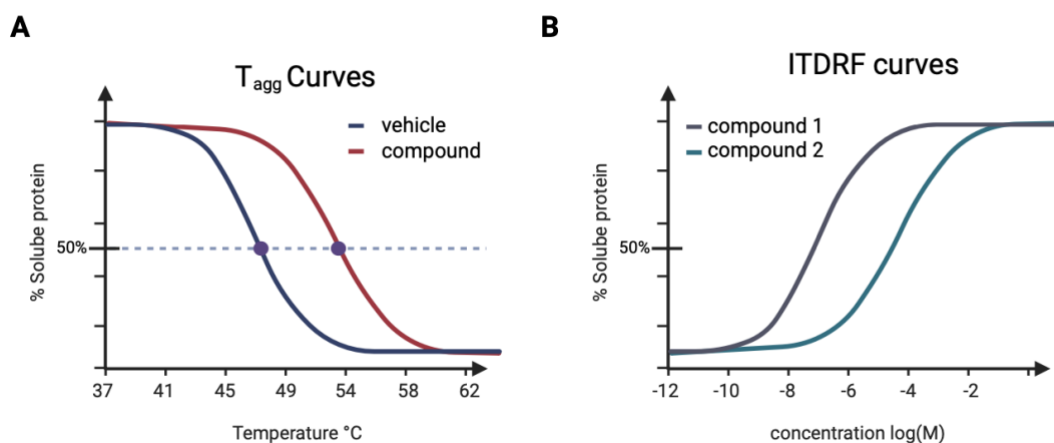


Figure 2. Example curves from CETSA experiments A) Tagg curves from a melt curve with the

Tagg marked by a purple dot on each curve. B) Example ITDR fingerprints (ITDRF). Created in BioRender. SEASHORE-LUDLOW, B. (2025) <https://BioRender.com/rw3yjkbb>

4.2 Procedure

CETSA experiments include the following general steps:

- 1) Compound treatment on cells or lysate
- 2) Heat challenge (transient heating)
- 3) Cell lysis (if using live cell compound treatment)
- 4) Detection

General considerations:

Detection Format

In general, it is easier to establish a Western blot CETSA, given the necessary antibodies or detection reagents are available.

Cell model

It is recommended to use as physiologically relevant cell systems as possible for the scale of experiments. CETSA in tissues and whole blood have been described^{2,12}.

Compound treatment

Compound treatment can be performed in lysates or in live cells. It is important to consider that in live-cell CETSA experiments, cellular processes, such as transport of the compound across the cell membrane and metabolic or post-translational modifications of the compound remain intact. This enables study of target binding in a complex, more physiologically relevant setting. In lysate it is possible to detect direct binders of a protein target.

For a further considerations when developing either a Western blot or AlphaLisa CETSA see related methods papers^{10,13}.

4.3 Controls

When looking for shifts in Tagg, vehicle should always be included as a control. In screening implementations, a known binder or ligand can be used as a positive control. If this is not available lysate or cells that have not been heat treated can be used as a positive control.

4.4 Information for consumables/ chemicals/ equipment

General instrumentation for Western blot CETSA:

- PCR machine for heat challenge, preferably with distinct temperature areas for the temperature series
- Centrifuge
- All materials and instrumentation for Western blot
 - Gel Electrophoresis Chamber
 - Gel Transfer system, e.g. iBlot

- Imaging or detection system

General instrumentation for AlphaLisa CETSA

- PCR machine with 96- or 384- well plate compatibility (eg ProFlex PCR)
- Liquid handler (eg MultiDrop Combi) or electronic pipettes
- Plate reader (eg EnVision, EnSight etc) for AlphaScreen or AlphaLisa readout or tagged protein readout

4.5 Reagent preparation/ calculation

Lysate

The most reliable method for generating lysate is several rounds of freeze thawing in a CETSA compatible buffer.

Live cell suspension

Cells are suspended in media at the required concentration depending on format.

Compound treatment

When screening with CETSA, as with other thermal shift assays, higher concentrations are recommended. This is because saturating conditions are often needed to observe a sufficiently large apparent shift in melting temperature¹⁰.

4.6 Measurements

For Western blot, gels are imaged and quantified using Fiji or similar software. These raw values are then plotted in Excel, GraphPad or similar.

For AlphaScreen or AlphaLisa plates are normalized to positive and negative controls.

A recent paper by Florez Weidinger et al details a data analysis workflow for screening data¹⁴.

5 Further documents

Literature references:

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6 Annex

In this section all documents are listed that are needed for the documentation.

- e.g. Annex 1: Revision